GEK-83751B

DRIVE
SYSTEM
INSTRUCTIONS

JOINT VOLTAGE CONTROL

3S7931JC100A5

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GENERAL ELECTRIC

ISSUE DATE: February 1981

REVISED: October 1981

REVISED: December 1986

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CHAPTER 1

1-1. INTRODUCTION

The 3S7931JClOO Joint Voltage Control provides a common voltage control for all units of a multi-unit generating station. Reactive load is balanced between machines in proportion to the rating of each machine.

1-1.1. RECEIVING AND HANDLING

Immediately upon receipt, the equipment should be carefully unpacked and examined for any damage that might have been sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the transportation company and the nearest General Electric Sales Office should be notified promptly.

1-1.2. DESCRIPTION

A one line diagram of the equipment is shown on sheet 1 of the elementary diagram. The joint voltage regulator has a motor driven voltage adjuster which is operated by supervisory control. Error output of the joint voltage regulator is applied to an indicating meter and a time delay relay which can be used for trip and/or alarm.

The voltage error is applied to a static motor drive for each regulator voltage adjuster. This is the principal motor drive input. Each drive positions its voltage adjuster to reduce the station voltage error to zero when all unit regulators have been repositioned.

Signals proportional to vars are obtained from var sensors connected to the instrument transformers associated with each machine. These signals are summed and fed to a total var indicator. Because the generators are identical, the proportioning var channel is simplified to an averaging channel. The sum is divided by the number of units to obtain a signal proportional to average vars. A var unbalance signal for each machine is obtained by comparing its own var signal with the average. This is a secondary error input to the static motor drive for each unit.

The joint voltage regulator circuit is shown in the top portion of sheet 5 of the elementary diagram. Input transformer lT has 5 volt taps from 75 to 145 volts for approximate match to the P.T. secondary. The rectified output of lT secondary is approximately 30 volts. Output voltage is nominally 0.1 volts per percent terminal voltage error. Potentiometers 7P and 8P control the range and level of the voltage adjuster.

The var transducer circuit is shown on the lower portion of sheet 5 of the elementary diagram. Separate instructions GEI-88846 cover the var transducer. (This publication can be ordered from the Instrument Dept. of General Electric Co., West Lynn, Mass.)

The var transducers are shown on elementary diagram sheet 2 and 5 together with amplifiers l-O.A. through 3-0.A. which amplify and filter the transducer outputs. The output of each amplifier is normalized by means of a feedback potentiometer. Test jacks 1-3TJ are provided for the amplifier outputs. The three var signals are summed by amplifier 7-0.A and total vars indicated by lM. The signals are averaged by 6-0.A. whose gain is varied in inverse proportion to the number of input signals. On the right side of sheet 2 are the motor drive input resistors. Current into each drive point is proportional to the algebraic sum of the inputs and opposite in sign. The inputs are unit vars, average vars, and the voltage error which has been amplified by 5-0.A. Note that this application is for generators of equal rating. Thus, the var proportioning operation simplifies to var averaging.

1-1.3. MOTOR CONTROL

The motor control circuitry is shown on sheet 10 of the elementary diagram. The input current signal, circuit 1, is summed with a feedback current from 21R, amplified and inverted by 5Q and 6Q. Added to the preamp output, circuit 10 to circuit 16, is an ac voltage which lags the ac supply by 90 degrees. A negative dc voltage is added to the circuit 16 voltage to obtain the forward firing-control voltage, circuit 17. A positive ac voltage is added to circuit 16 voltage to obtain the reverse firing control voltage, circuit 15.

The relaxation oscillator lUJT, 8R and 4C fires the forward thyristor with pulses of current through the primary of 2T if 2Q is non-conducting. This requires that lQ be conducting and circuit 17 be positive. The reverse firing circuit oscillates if 4Q is non-conducting. This requires that 3Q be non-conducting and circuit 15 be negative. The dc offset in circuit 15 and 17 voltages can be varied by lP to control the zero input firing point of the thyristors. A negative input signal, inverted by the preamp, raises circuit 15 and 17 voltages equally, causing forward firing to occur earlier in the cycle and reverse firing to occur later or not at all, depending upon whether circuit 15 voltage goes negative or remains positive. A current limit circuit prohibits full voltage from being applied to a motor at standstill. An ac voltage lagging the supply by 90 degrees is obtained from BC and 15R. It is combined with the back emf of the motor by 16R ad 17R and fed through 18R, 5D and 7D to the base of firing control transistors lQ and 3Q. The effect is to prohibit firing until the resultant current will not exceed a safe value.

WARNING

This equipment contains a potential hazard of electrical

shock or burn. Only personnel who are adequately trained and thoroughly familiar with the equipment and these instructions should install, operate, or maintain this equipment.

1-1.4. TEST AND ADJUSTMENT

WARNING

This equipment is designed to operate in a normal industrial atmosphere. Operating radio transmitters nearby may cause the control equipment to misoperate. It is recommended that radio transmitters not be operated within the vicinity of the control equipment when the panel doors are open, nor at any time within the confines of any power control room.

WARNING

Isolation of test equipment from the equipment under test presents potential electrical hazards. If the test equipment cannot be grounded to the equipment under test, the test equipment's case must be shielded to prevent contact by personnel.

1-1.5. Joint Voltage Regulator

1. Set 5P input gain control of 5-0.A at midpoint.
2. Adjust taps on lT to nearest 5 volts corresponding to nominal P.T. voltage.
3. Insert a variable autotransformer in place of the signal PT and adjust 7P and 8P for the desired voltage range and maximum voltage. With the joint voltage control in TEST position, operate lOM to move lOP wiper to the maximum voltage end. Set the autotransformer for the desired maximum voltage and adjust 8P to obtain 0 output from 5-0.A. It is recommended that an oscilloscope be used in adjustment of this equipment. Set the input voltage at the desired minimum, the voltage adjuster lOP at the other end of its travel and adjust 7P for 0 output from 5-0.A. Repeat the adjustment of 8P and 7P until no further adjustment is required. Check meter 2M for proper indication.

1-1.6. VAR Transducers and Amplifiers

1. Apply nominal PT voltage to each var transducer. Place an in-phase current into the current input. The value of this current should be the reactive CT secondary current at rated MvA and p.f. Adjust the gain of each var signal amplifier to obtain +5.0 volts output.
2. Calibrate the total var meter by means of 6P gain control for amplifier 7-0.A. See notes at the bottom of sheet 2 of the elementary diagram.

1-1.7. Unit Motor Drives

The motor control boards are factory adjusted for optimum operation and further adjustment should be required only if component parts are replaced. Adjustment is made with the motor connected to the motor control. Unplug amplifiers l-O.A. through 6-0.A. and ground vTJ test point. An extender board should be inserted between the motor control board and its connector. Connect an oscilloscope to the motor armature being careful to connect the oscilloscope ground to the motor ground. Apply power and rotate the balance adjustment 2P (nearest pin end of board) to stop the motor. This will be indicated on the oscilloscope by a trace of which the horizontal portion is zero volts, average, indicating 0 Vdc back emf from the motor. The overlap adjustment lP (nearest handle end of board) is set to obtain a horizontal trace 2 1/2 times the length of other portion of the trace in which the thyristor is conducting; this corresponds to approximately 40 degree overlap.

1-1.8. Level Sensor

With 0 volts at input pin 35, 6P should be adjusted for 0 volts at BT. P.; the relay connected to output pin 31 should be energized. It should drop out at the desired positive-going or negative-going level set by 16P (external) and/or 4P and 5P.

1-1.9. Pushbuttons, JVC Switch, Relays, and Lights

These components should function to place generating units 1, 2, and 3 in joint voltage control and indicate their status as shown on the elementary diagram. Relays and lights operate by 24 volts from the power supply, or from 125 volts dc (except for the excessive voltage error relay 60).

1-1.10. Power Supply

All ac and dc voltages shown on the elementary diagram should be measured periodically.

1-1.11. Motor Operated Voltage Adjuster

The gearmotor speed should be set at 3/4 r.p.m. with resistor Rl. Check for Proper relay operation, potentiometer direction of rotation, and cam-operated switches.

1-1.12. Cabinet Light, Receptacles and Heater

These require little maintenance or adjustment.

WARNING

TO PREVENT HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MOST BE ADHERED TO.

1-2. OPERATING INSTRUCTIONS

The function of the joint voltage regulator and var controller is to replace operators in changing the voltage set-points of the several generating units in the station as power system demand varies. It is not a fast-acting regulator in itself, but it will continuously adjust the fast-acting voltage regulators in an effort to maintain constant station output voltage and proper reactive power division.

The operational amplifier networks, meter calibration, and motor-drive units are factory preset as indicated on the elementary diagram. The voltage regulator range and level rheostats may require field adjustment - perhaps even the regulator input transformer tap connection may require changing. The level sensor threshold can be changed after gaining operating experience which may indicate a more desirable setting. Also the time delay of the relay which it operates may be changed.

The operating controls are the local and remote control switches. Relays 90JIX2-1, -2, and -3 place their respective generators under joint control. Those relays are operator selected, with unit line circuit breaker interlocking. The selector switches, remotely located, are CS-I/JVC, CS-2/JVC, and CS-3/JvC. Thus a generating unit must be connected to the power system in order to be put under joint control. If a generating unit's excitation system is transferred to dc regulator control, it should be removed from joint voltage control. Also, if a unit's var-sensing circuitry goes out of calibration, it should be removed from joint control. Loss of station battery control power, or trouble with the internal power supply will automatically take the joint control out of service. If two of the three units must be removed from joint voltage control, the remaining unit should be removed, also.

Control switch ICS and pushbutton switches IPB, 2PB and 3PB may be used to operate the test relays 90JIXI-I, -2 and -3, which are provided to test each of the three channels in the equipment without actually placing that channel in service.

The joint regulator set-point can be changed, remotely, by IOCS.

1-2.1. REFERENCE DOCUMENTS

44C308284	Elementary Diagram
44B309216	Outline Drawing
GEI-93161	Reversing Statotrol* Drives, 3S7507RS series
GEI-88846	Watt Transducer
278A2011	Factory Test Instructions

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1-3. RENEWAL PARTS

When ordering renewal parts, address the nearest General Electric Sales Office; specify the quantity required, and circuit #ymbol as shown on the elementary diagram. In addition, give the 3S number of the equipment. The principal Electrical parts in the equipment are given in the following tabulation:

DESIGNATION	QTY.	PART No.	DESCRIPTION
LSW	1	44C372098-G02	Switch, Receptacles
HTR	1	SD2082	Space Heater
23	1	6A426GI	Thermostat
HSW	1	TEB122015	Circuit Breaker
ST	1	206B4414P5	Transformer
QID-Q6D	6	68A7250P2	Diode
QIC, Q2C, Q3C	3	68A7051P0470F	Capacitor,.047 MFD.
IR-I,IR-2, IR-3	3	68A7004B50E390JA/XIOIAE	Resistor, Adj., 390 ohms, 50W
IMCR, 2MCR, 3MCR	3	68A7004A20CIOOOCA	Resistor, IK, 20W
IFU - 4FU	4	CSC#OT6	Fuse, 6A
LTG	1	167A8499P2	Lamp, 220V, 50W
IM	1	50-162012MTMT2LLH	VAR. Meter (DCVM.)
2M	1	933B574P15	Volt. Error Mtr. (DCVM.)
1-4TJ, ATJ, GTJ	6	289A700P2	Jack
ICS	1	CR2940UB203B	Selector Switch

DESIGNATION	QTY.	PART No.	DESCRIPTION
1-3PB	3	CR103DC221B	Pushbutton Switch
ICS/IL	1	CR103DD221X	Indicator
	8	125A4423P28	Lamp, 28V
Var. Trans. 1, 2, 3	3	50-470121AWAAI	Watt Transducer
	3	STA #SJK-20K04-S	Transducer CT SWitch
60X	1	CR2820B211AA41K	Time Delay Relay, 125V
	1	3S7932HA115G58	Motor-Op. Voltage Adjuster
IOM	1	278A3369PI	Motor, P.M., 75V
K	1	218A4274P21	Relay, 125V
IOP	1	573A384P15A	Potentiometer, 5K, 25W
RI	1	B50E2500JA/XIOIAE	Resistor, Adj. 2.5K,50W
LS1-6	6	44A212115-001	Cam-Operated Switch
	1	3S7931JC200AII	Relay Module
4JV, 27V, 90JIX-(6)	10/8	3S2791G138A3	Relay, Min., 125V
27A, 27B, 60	5/3	3S2791G138A4	Relay, Min., 28V
	8/6	68A7250P2	Diode, IA.
	1	3S7931JC200A12	Power Supply Module
3Z, 5Z	2	68A7233P07BN	Zener Diode, 12v
6Z	1	68A8210P0621	Zener Diode, 6V
4Z	1	44A213761-001	Zener Diode, 5V
3T	1	206B4414PI	Transformer
IX, 2X	2	44B211629-GOI	Reactor
ICB	1	CRIOIH	Switch
	1	CR123H12.2B	Overload Heater for ICB
IC, 2C	2	44A339814-M15	Capacitor, EL.,1500 MFD, 50 V
IR, 2R	2	68A7004A20C22DA/XIOIAC	Resistor, 22 ohm, 20W
IZ, 2Z	2	44A337972-005	Zener Diode, 15V
IT	1	44B213349-GOI	Transformer
2T	1	395B581PI	Transformer
1-4D	4	167A8419PI	Diode
	1	44B315610-G03	Component Board
5D	1	68A7250P2	Rect. IA 500V
7R,8R	2	68A7004AIOBIOOCA	Res., 100 Ohm, IOW
5R,6R	2	68A7004AIOB68CA	Res. 680hms, IOW
3R	1	68A7001P471	Resistor, 470 Ohm, IW
4R	1	68A7001P102	Resistor, 1000 Ohm, IW
10, IIR	2	68A7004AIOB2200CA	Res., 2200 Ohm
9RI, 9R2	2	68A7004AIOB33CA	Resistor, 33 Ohm, IOW
	1	3S7931JC200A4	Voltage Regulator
IT	1	44B216024-GOI	Transformer
IX	1	44B211608-GOI	Reactor
IZ	1	IN2620B	Zener Diode, 9.3V
1-4D	4	44B232011-001	Diode
3RA	1	687001P471	Resistor, 470 Ohm, IW
IR	1	68A7001P331	Resistor, 330 Ohm, IW
2R	1	6BA7001P221	Resistor, 220 ohm, IW
3RB	1	68A7001P151	Resistor, 150 Ohm, IW
7P,8P	2	277A7704PI	Potentiometer, 250 Ohm, 2W
IC	1	44A339814-LIO	Capacitor, 100 MFD, 50V
	4/3	44D236353-G02	Motor Control Print Ckt. Bd.
13R, 14R	2	277A7785C47	Res. 4.7 Ohms, 3W
IUJT, 2UJT	2	68A7369PI	Transistor
2T, 3T	2	933B272PI	Transformer
15R	1	68A7001P561	Res. 560 Ohms, IW
17R	1	68A7001P222	Res. 2.2K Ohms, IW

DESIGNATION	QTY.	PART No.	DESCRIPTION
lC	1	44A337922-J20	Capacitor, 2 MFD.
lP	1	277A7782F50	Pot 5K Ohms, .5W
2P	1	277A7782G50	Potentiometer, 50K, .5W
9C	1	44A337973-GIO	Cap. .01 UF, 500 VDC
lCD, 2CD	2	44B212739-007	Thyristor
lT	1	44B311324-GOl	Transformer
30R, 31R	2	68A7000P471	Res. 470 Ohms, .5W
4R, 5R, 9R	3	68A7000P102	Resistor, lK, .5W
lR	1	68A7000P122	Res. 1.2K Ohms .5W
8R, 12R	2	68A7000P332	Res. 3.3K Ohms .5W
18R, 27R	2	68A7000P472	Res. 4.7K Ohms .5W
23R	1	68A7000P682	Res. 6.8K Ohms .5W
28R	1	68A7000P103	Res. lOK Ohms .5W
26R	1	68A7000P243	Res. 24K Ohms .5W
6R, lOR, 22R	3	68A7000P273	Res. 27K Ohms .5W
32R	1	68A7000P513	Res. 51K Ohms .5W
7R	1	68A7000P563	Res. 56K Ohms .5W
24R	1	68A7000P154	Res. .15 MegOhms .5W
llR	1	68A7000P224	Res. .22 MegOhms .5W
16R	1	68A7001P562	Res. 5.6K Ohms lW
25R	1	68A7001P753	Res. 75K Ohms lW
4C, 5C, llC	3	206B4404H22	Capacitor, .22 MFD.
8C	1	44A337922-J40	Capacitor, 4 MFD.
2C, 3C	2	206B4436L50	Capacitor 500MFD 12V
6C, 7C	2	278A2115MlO	Capacitor, 1000 MFD, 25V
lD-12D, 19D-24D	18	44B232011-001	Diode, 100 MA.
4Q	1	277A7793P7	Transistor
lQ, 2Q, 3Q, 5Q, 6Q	5	206B4429-001	Transistor
21R	1	68A7000P134	Res. 130K Ohms .5W
lO.A.-80.A	5/3	44D236354-G03	Stabilized Op. Ampl. Printed Circuit Board
lA, 2A	2	278A2110-001	OP. Amplifier
lP, 2P	2	983B583H20	Pot., 200K Ohms .25W
lD-4D	4	206B4422Pl	Diode
lZ-4Z	4	44A376684-C91	Zener 9.1V 400MW
lR-4R	4	68A7000P183	Res. 18K Ohms .5W
5R-6R		2 68A7000P105	Res. l.OM Ohms .5W
lC,2C	2		Capacitor, 330 pf.
	2/1	IC3606JVCAlA	Resistor Printed Ckt. Bd.
lR	1	68A7030P562F	Res 56.2K Ohms
2R	1	68A7030PlOOE	Res. lK Ohm
102R, 202R			
302R, 104R			
204R, 304R			
105R, 205R			
305R	9	68A7030PlOOF	Res. lOK Ohms
lOlR, 201R	3	68A7030P221E	Res. 2.21K Ohms
301R			
106R, 107R			
206R, 207R			
306R, 307R	6	68A7030P121E	Res. 1.21K Ohms
108R, 208R			
103R, 203R			
303R	5	68A7030P499E	Res. 4.99K Ohms
308R	1	68A7030P750E	Res. 7.50K Ohms
lP-3P, 5P, 6P	5	68A7125P502	Pot. 5K Ohms

DESIGNATION	QTY.	PART No.	DESCRIPTION
9P, 12P	2	68A7125P103	Pot. IOK Ohms
16P	1	68A7125P253	Pot. 25K Ohms
IOIC, 201C, 301C	3	68A7052PIOHIO	Cap. 10 uf
IC	1	68A7052P4A20	Cap. 4 uf
LS-2	2/1	44D221533-GOI	Level Senser Printed Ckt. Bd.
ID-4D, 9D-12D	8	IN459	Diode 40MA 200V
IBD, 2BDS	2	IN3020B	Zener Diode IOV IW
IQ, 2Q, 5Q, 6Q	4	44B311057-001	Transistor
3P, 6P	2	983B558F50	Pot. 5K Ohms, 1/4 W.
IP, 2P, 4P, 5P	4	983B558GIO	Pot. IOK Ohms, 1/4 W.
3Q, 4Q, 7Q, BQ	4	44B238203-001	Transistor
5D-8D, 13D-16D	8	44B232011-001	Diode
IC, 2C	2	933B597G47	Capacitor .047 MFD
IIOR, IIR			
25R, 26R	4	933B590D47	Res. 47 Ohms, 1/2 W.
9R, 24R	2	933B590F33	Res. 3.3K Ohms, 1/2 W.
3R-5R, 18R-20R	6	933B590F47	Res. 4.7K Ohms, 1/2 W.
7R, 22R	2	933B590F56	Res. 5.6K Ohms, 1/2 W.
15R, 27R	2	933B590F68	Res. 6.8K Ohms, 1/2 W.
13R, 30R	2	933B590G12	Res. 12K Ohms, 1/2 W.
6R, 21R	2	933B590G18	Res. 18K Ohms, 1/2 W.
2R, 8R, 17R, 23R	4	933B590HIO	Res. IOOK ohms, 1/2 W.
12R, 29R	2	933B590H22	Res. 220K Ohms, 1/2 W.
14R, 28R	2	M6986159F18	Res. 1.8K Ohms, 1/2 W.
IR	1	M6936159EIO	Res. IOOK Ohms, 1/2 W.
16R	1	M6986159E47	Res. 470K ohms, 1/2 W.
	1	44D222911-GOI	Extender Board
	1	3S7932HA115G58	Motor-operated voltage adj.
IOM	1	278A3369PI	Motor
K	1	218A4274P21	Relay
RI	1	68A7004B25D3600DA/XIOIADD	Resistor, 3.6K,25 W.
IOP	1	573A384P15A	Potentiometer, 5K,25W.
LSI-LS6	6	44A212115-001	Switch
	6	44A217958-GOI	Cam Assembly
	1	44A217974-002	Coupling
	1	44B314828-002	Coupling Body
	2	44B314828-003	Coupling Segment
	1	44A217945-001	Shaft
IOCS			
10 P Lights			

GENERAL ELECTRIC

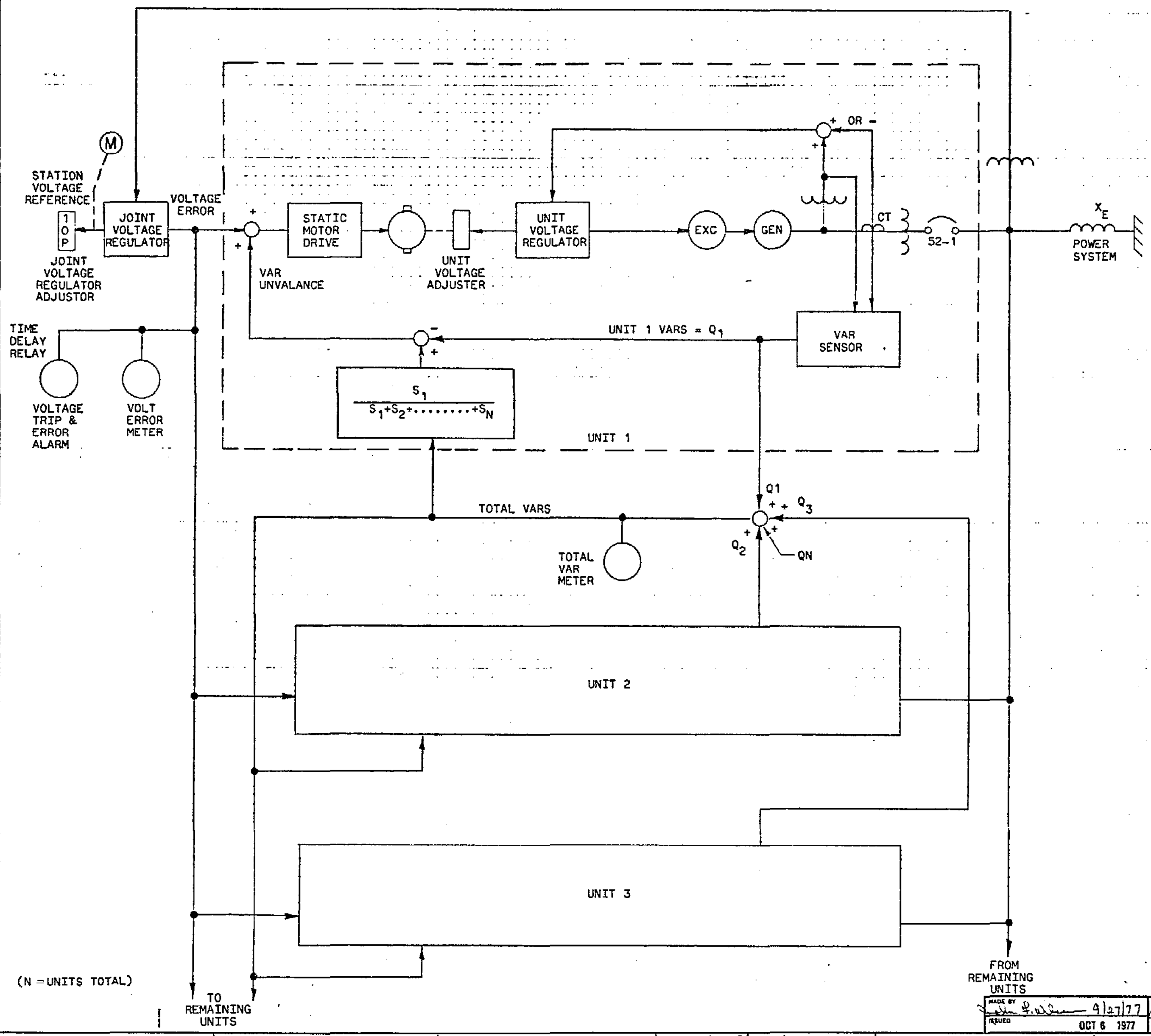
44C308284
CONT ON SHEET 2 SH NO. 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:–
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± DECIMALS ± ANGLES ±

TITLE
JOINT VOLTAGE CONTROL
FIRST MADE FOR 3S7931JC100A5
ELEMENTARY DIAGRAM

PALMAR – URUGUAI
REQN. 140-J11-T80284

ONE LINE DIAGRAM



M
STATION VOLTAGE REFERENCE
1 O P
JOINT VOLTAGE REGULATOR ADJUSTOR
JOINT VOLTAGE REGULATOR
VOLTAGE ERROR
TIME DELAY RELAY
VOLTAGE TRIP & ERROR ALARM
VOLT ERROR METER
STATIC MOTOR DRIVE
VAR UNVALANCE
UNIT VOLTAGE ADJUSTER
UNIT VOLTAGE REGULATOR
EXC
GEN
+ OR –
CT
52-1
X_E
POWER SYSTEM
VAR SENSOR
UNIT 1 VARS = Q_1
$\frac{S_1}{S_1+S_2+\cdots\cdots+S_N}$
UNIT 1
TOTAL VARS
TOTAL VAR METER
Q1
Q_2
Q_3
QN
UNIT 2
UNIT 3
(N = UNITS TOTAL)
TO REMAINING UNITS
FROM REMAINING UNITS

1CS – JVC TEST SWITCH
1PB – JVC TEST PUSHBUTTON FOR GEN. NO. 1
2PB – JVC TEST PUSHBUTTON FOR GEN. NO. 2
3PB – JVC TEST PUSHBUTTON FOR GEN. NO. 3
4JV – APPLIES 120 VAC INPUT TO POWER SUPPLY
27A – POSITIVE 24 VOLT MONITORING RELAY
27B – NEGATIVE 24 VOLT MONITORING RELAY
27JV – AUXILIARY RELAY TO 27A AND 27B ENABLES JVC OPERATION WHEN POWER SUPPLY IS ENERGIZED
60 – VOLTAGE ERROR SENSING RELAY
60X – VOLTAGE ERROR ALARM RELAY
90J1X1-1 – APPLIES VAR SIGNAL FROM GEN. NO. 1 TO SUMMING AMPLIFIERS AND MOTOR DRIVE CIRCUIT, ETC.
90J1X1-2 – APPLIES VAR SIGNAL FROM GEN. NO. 2 TO SUMMING AMPLIFIERS AND MOTOR DRIVE CIRCUIT, ETC.
90J1X2-1 – JVC TRANSFER RELAY FOR GEN. NO. 1 TRANSFER CONTROL OF VOLTAGE-ADJUSTING POTENTIOMETER FROM OPERATORS SWITCH TO JVC ETC.
90J1X2-2 – JVC TRANSFER RELAY FOR GEN. NO. 2 TRANSFER CONTROL OF VOLTAGE-ADJUSTING POTENTIOMETER FROM OPERATORS SWITCH TO JVC, ETC.
52 – GENERATOR CIRCUIT BREAKER

NOTES:
** ITEMS SUPPLIED BY CUSTOMER
* ITEMS SUPPLIED BY OTHER DEPTS.
◎ (Q NO'S.) OUTGOING CONNECTION POINTS FROM JVC CUBICLE.
△ MOUNTED ON CUSTOMERS CONTROL BOARD.
o PANEL TERMINAL POINTS IN JVC CUBICLE
▣ CUSTOMER CONNECTION POINTS (IN OTHER EQUIPMENT FROM JVC CUBICLE)
S = VAR RATING FOR EACH GENERATOR
(S_1 = VAR RATING FOR UNIT 1,
S_2 = VAR RATING FOR UNIT 2, ETC.)

REVISIONS		PRINTS TO	
1	REVISED [illegible] McLaughlin APP. 12 1973		3S4.

MADE BY [illegible] 9/27/77
ISSUED OCT 6 1977
APPROVALS [illegible]
DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT
WAYNESBORO, VA. LOCATION
44C308284
CONT ON SHEET 2 SH NO. 1

PF-809-F (2-77)
PRINTED IN U.S.A.

GENERAL ELECTRIC

44C308284 CONT ON SHEET 3 SH NO. 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—

APPLIED PRACTICES	SURFACES	FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

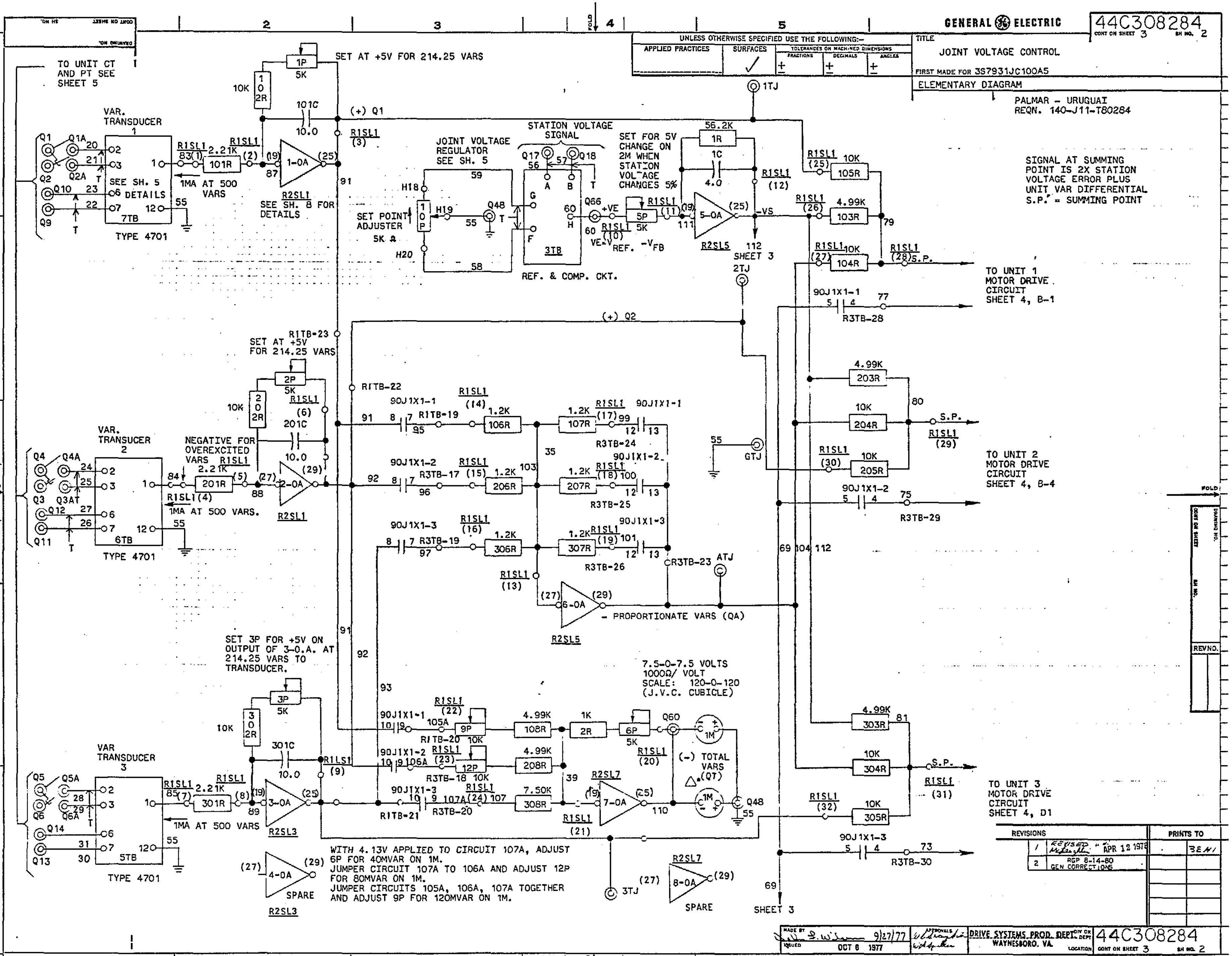
TITLE

JOINT VOLTAGE CONTROL

FIRST MADE FOR 3S7931JC100A5

ELEMENTARY DIAGRAM

PALMAR – URUGUAI
REQN. 140-J11-T80284


TO UNIT CT AND PT SEE SHEET 5
VAR. TRANSDUCER 1
TYPE 4701
7TB
SEE SH. 5 DETAILS
Q1 Q1A Q2 Q2A Q10 Q9
20 21 22 23
1MA AT 500 VARS
R1SL1 83(1) 2.21K 101R
R1SL1 (2)
1P 5K
SET AT +5V FOR 214.25 VARS
10K 1 0 2R
101C 10.0
(+) Q1
1-0A (19) (25) 87
R2SL1
SEE SH. 8 FOR DETAILS
R1SL1 (3)
91
JOINT VOLTAGE REGULATOR SEE SH. 5
H18 H19 H20
SET POINT ADJUSTER 5K
1 0 P
Q48 T 55
59 58
STATION VOLTAGE SIGNAL
Q17 Q18 56 57
A B G F T
60 H
3TB
REF. & COMP. CKT.
Q66 +VE
60 VE-
R1SL1 (10)
REF. -VFB
5P 5K
R1SL1 (11)
SET FOR 5V CHANGE ON 2M WHEN STATION VOLTAGE CHANGES 5%
56.2K 1R
1C 4.0
5-0A (19) (25) 111
R2SL5
R1SL1 (12)
-VS
112 SHEET 3
1TJ
2TJ
R1SL1 (25) 10K 105R
R1SL1 (26) 4.99K 103R
79
R1SL1 (27) 10K 104R
R1SL1 (28) S.P.
TO UNIT 1 MOTOR DRIVE CIRCUIT SHEET 4, B-1
90J1X1-1 5 4 77
R3TB-28
(+) Q2
R1TB-23
SET AT +5V FOR 214.25 VARS
2P 5K
10K 2 0 2R
R1SL1 (6)
201C 10.0
R1TB-22
VAR. TRANSUCER 2
TYPE 4701
6TB
Q4 Q4A Q3 Q3AT Q12 Q11
24 25 26 27
NEGATIVE FOR OVEREXCITED VARS
R1SL1 84 2.21K 201R (5)
R1SL1 (4)
1MA AT 500 VARS.
2-0A (27) (29) 88
R2SL1
91 92
90J1X1-1 8 7 R1TB-19 95
R1SL1 (14) 1.2K 106R
1.2K 107R
R1SL1 (17) 99
90J1X1-1 12 13
R3TB-24
90J1X1-2 8 7 R3TB-17 96
R1SL1 (15) 1.2K 206R
103 35
1.2K 207R
R1SL1 (18) 100
90J1X1-2 12 13
R3TB-25
90J1X1-3 8 7 R3TB-19 97
R1SL1 (16) 1.2K 306R
1.2K 307R
R1SL1 (19) 101
90J1X1-3 12 13
R3TB-26
R3TB-23
ATJ
R1SL1 (13)
6-0A (27) (29)
- PROPORTIONATE VARS (QA)
R2SL5
55 GTJ
4.99K 203R
80
10K 204R
S.P.
R1SL1 (29)
R1SL1 (30) 10K 205R
TO UNIT 2 MOTOR DRIVE CIRCUIT SHEET 4, B-4
90J1X1-2 5 4 75
R3TB-29
69 104 112
SET 3P FOR +5V ON OUTPUT OF 3-O.A. AT 214.25 VARS TO TRANSDUCER.
3P 5K
10K 3 0 2R
301C 10.0
R1LS1 (9)
VAR TRANSDUCER 3
TYPE 4701
5TB
Q5 Q5A Q6 Q6A Q14 Q13
28 29 30 31
R1SL1 85(7) 2.21K 301R
R1SL1 (8)
1MA AT 500 VARS
3-0A (19) (25) 89
R2SL3
4-0A (27) (29)
SPARE
R2SL3
93
90J1X1-1 10 9 105A
R1SL1 (22) 9P
R1TB-20 10K
4.99K 108R
90J1X1-2 10 9 106A
R1SL1 (23) 12P
R3TB-18 10K
4.99K 208R
90J1X1-3 10 9 107A
R1SL1 (24) 107
R3TB-20
R1TB-21
7.50K 308R
1K 2R
6P 5K
R1SL1 (20)
39
R2SL7
7-0A (19) (25) 110
R1SL1 (21)
Q60
1M
(-) TOTAL VARS (QT)
Q48 55
7.5-0-7.5 VOLTS 1000Ω/ VOLT SCALE: 120-0-120 (J.V.C. CUBICLE)
3TJ
R2SL7
8-0A (27) (29)
SPARE
4.99K 303R
81
10K 304R
S.P.
R1SL1 (31)
R1SL1 (32) 10K 305R
TO UNIT 3 MOTOR DRIVE CIRCUIT SHEET 4, D1
90J1X1-3 5 4 73
R3TB-30
69 SHEET 3
A B C D E
1 2 3 4 5 6
FOLD

SIGNAL AT SUMMING POINT IS 2X STATION VOLTAGE ERROR PLUS UNIT VAR DIFFERENTIAL
S.P. = SUMMING POINT

WITH 4.13V APPLIED TO CIRCUIT 107A, ADJUST 6P FOR 40MVAR ON 1M.
JUMPER CIRCUIT 107A TO 106A AND ADJUST 12P FOR 80MVAR ON 1M.
JUMPER CIRCUITS 105A, 106A, 107A TOGETHER AND ADJUST 9P FOR 120MVAR ON 1M.

DRAWING NO.
CONT ON SHEET SH NO.
REV NO.

REVISIONS

1	REVISED APR 12 1978	
2	RGP 8-14-80 GEN CORRECTIONS	

PRINTS TO
BEN/

MADE BY 9/27/77
ISSUED OCT 6 1977

DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.

44C308284 CONT ON SHEET 3 SH NO. 2

FF.808-F (2-77)
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GENERAL ELECTRIC 44C308284 CONT ON SHEET 4 SH NO. 3

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:–

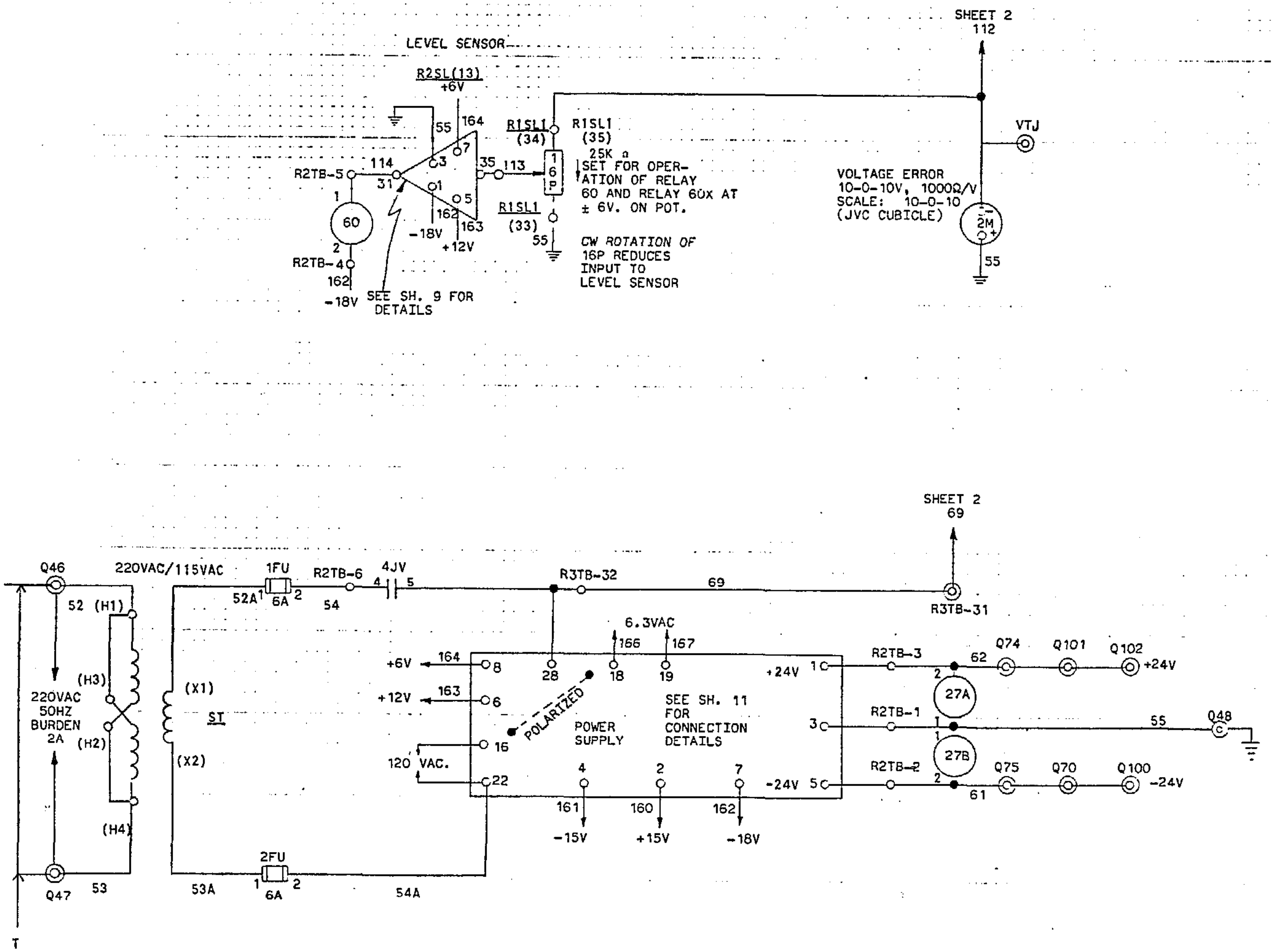
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE JOINT VOLTAGE CONTROL

FIRST MADE FOR 3S7931JC100A5

ELEMENTARY DIAGRAM

PALMAR – URUGUAI
REQN. 140-J11-T80284



LEVEL SENSOR
R2SL(13)
+6V
55
164
R1SL1 (34)
R1SL1 (35)
25K Ω
SET FOR OPERATION OF RELAY 60 AND RELAY 60X AT ± 6V. ON POT.
R2TB-5
114
31
3
7
1
5
162
163
35
113
1
6
P
R1SL1 (33)
55
CW ROTATION OF 16P REDUCES INPUT TO LEVEL SENSOR
-18V
+12V
60
R2TB-4
162
-18V
SEE SH. 9 FOR DETAILS
SHEET 2
112
VTJ
VOLTAGE ERROR
10-0-10V, 1000Ω/V
SCALE: 10-0-10
(JVC CUBICLE)
2M
55
SHEET 2
69
Q46
220VAC/115VAC
1FU
6A
52A
R2TB-6
54
4JV
4
5
R3TB-32
69
R3TB-31
52 (H1)
(H3)
(H2)
(H4)
220VAC
50HZ
BURDEN
2A
(X1)
ST
(X2)
6.3VAC
166
167
+6V
164
8
28
18
19
+24V
1
+12V
163
6
POLARIZED
POWER SUPPLY
SEE SH. 11 FOR CONNECTION DETAILS
3
16
120 VAC.
22
4
2
7
-24V
5
161
160
162
-15V
+15V
-18V
R2TB-3
62
Q74
Q101
Q102
+24V
27A
R2TB-1
55
Q48
27B
R2TB-2
Q75
Q70
Q100
-24V
61
Q47
53
53A
2FU
6A
54A
T

REVISIONS

		PRINTS TO
1	REVISED APR 12 1978	3EH1
2	RGP 8-14-80 GEN CORRECTIONS	

MADE BY D. Wilson 9/27/77
ISSUED OCT 6 1977
APPROVALS

DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT
WAYNESBORO, VA. LOCATION

44C308284
CONT ON SHEET 4 SH NO. 3

PF-509-F (2-77)
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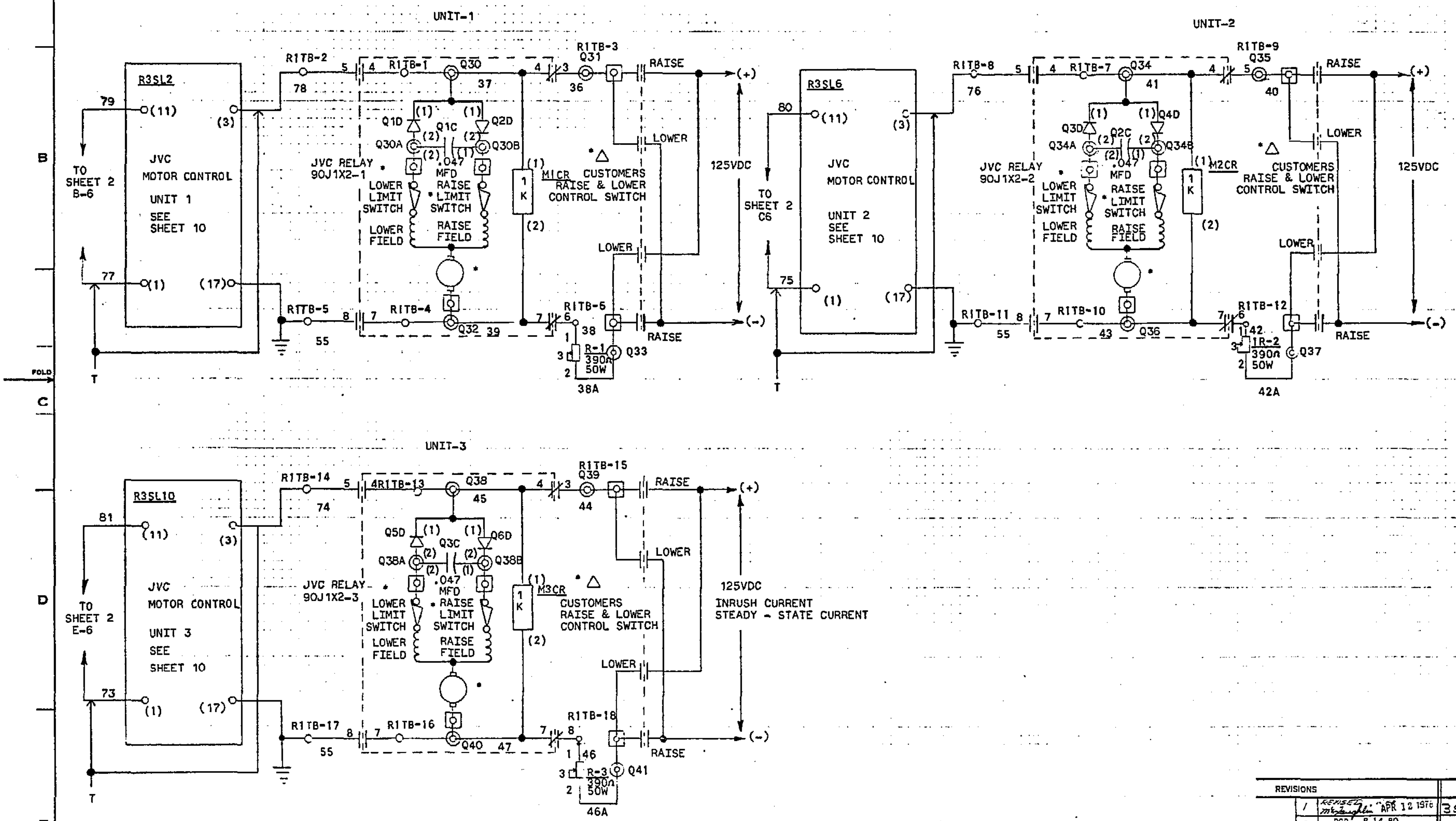
GENERAL ELECTRIC

44C308284
CONT ON SHEET 5 SH NO. 4

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:-

APPLIED PRACTICES	SURFACES	FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE JOINT VOLTAGE CONTROL
FIRST MADE FOR 3S7931JC100A5
ELEMENTARY DIAGRAM
PALMAR - URUGUAI
REQN. 140-J11-T80284



UNIT-1
R3SL2
JVC MOTOR CONTROL UNIT 1 SEE SHEET 10
(11)
(3)
(1)
(17)
79
77
TO SHEET 2 B-6
T
R1TB-2
78
R1TB-1
Q30
37
R1TB-3
Q31
36
RAISE
LOWER
(+)
(-)
125VDC
Q1D
Q1C
Q2D
Q30A
Q30B
.047 MFD
JVC RELAY 90J1X2-1
LOWER LIMIT SWITCH
RAISE LIMIT SWITCH
LOWER FIELD
RAISE FIELD
1 K
M1CR
CUSTOMERS RAISE & LOWER CONTROL SWITCH
R1TB-5
55
R1TB-4
Q32
39
R1TB-6
38
R-1 390Ω 50W
Q33
38A
UNIT-2
R3SL6
JVC MOTOR CONTROL UNIT 2 SEE SHEET 10
80
75
TO SHEET 2 C6
R1TB-8
76
R1TB-7
Q34
41
R1TB-9
Q35
40
Q3D
Q2C
Q4D
Q34A
Q34B
JVC RELAY 90J1X2-2
M2CR
R1TB-11
55
R1TB-10
43
Q36
R1TB-12
42
1R-2 390Ω 50W
Q37
42A
UNIT-3
R3SL10
JVC MOTOR CONTROL UNIT 3 SEE SHEET 10
81
73
TO SHEET 2 E-6
R1TB-14
74
R1TB-13
Q38
45
R1TB-15
Q39
44
Q5D
Q3C
Q6D
Q38A
Q38B
JVC RELAY 90J1X2-3
M3CR
125VDC INRUSH CURRENT STEADY - STATE CURRENT
R1TB-17
55
R1TB-16
Q40
47
R1TB-18
46
R-3 390Ω 50W
Q41
46A

REVISIONS		PRINTS TO
1	REVISED McLaughlin APR 12 1978	3EM
2	RGP 8-14-80 GEN CORRECTIONS	

MADE BY F. W. Wilson 9/27/77
ISSUED OCT 6 1977
APPROVALS
DRIVE SYSTEMS PROD. DEPT. WAYNESBORO, VA.
44C308284
CONT ON SHEET 5 SH NO. 4

FF-609-F (8-77) PRINTED IN U.S.A.

GENERAL ELECTRIC

44C308284
CONT ON SHEET 6 SH NO. 5

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:–

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
JOINT VOLTAGE CONTROL

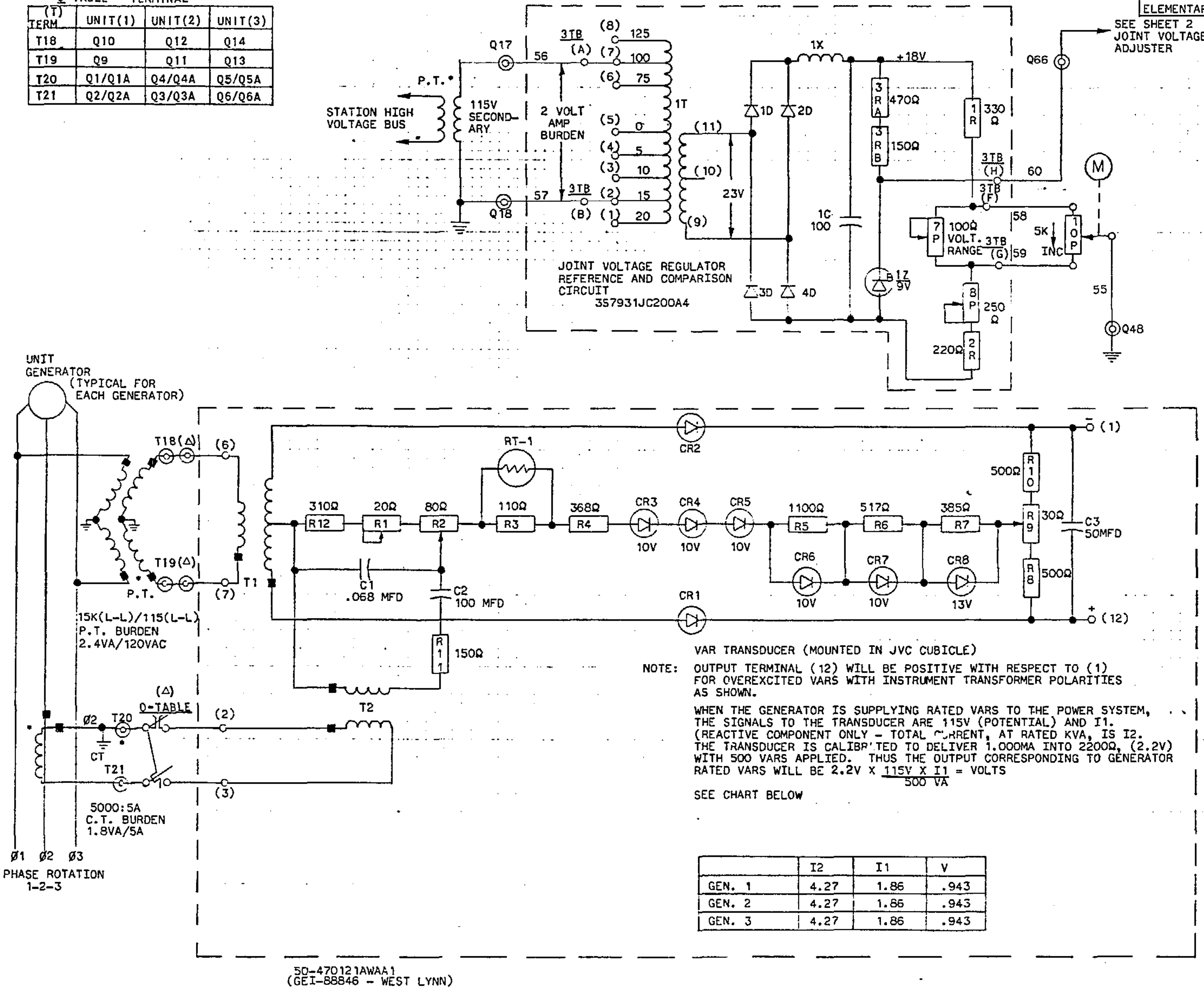
FIRST MADE FOR 3S7931JC100A5

ELEMENTARY DIAGRAM

PALMAR – URUGUA1
REQN. 140-J11-T80284

Q-TABLE – TERMINAL

(T) TERM	UNIT(1)	UNIT(2)	UNIT(3)
T18	Q10	Q12	Q14
T19	Q9	Q11	Q13
T20	Q1/Q1A	Q4/Q4A	Q5/Q5A
T21	Q2/Q2A	Q3/Q3A	Q6/Q6A


SEE SHEET 2
JOINT VOLTAGE
ADJUSTER
Q66
Q17
Q18
Q48
56
57
60
55
58
59
P.T.
STATION HIGH VOLTAGE BUS
115V SECONDARY
2 VOLT AMP BURDEN
3TB (A)
3TB (B)
(8) 125
(7) 100
(6) 75
(5) 0
(4) 5
(3) 10
(2) 15
(1) 20
1T
(11)
(10)
(9)
23V
1X
1D
2D
3D
4D
1C 100
+18V
3RA 470Ω
3RB 150Ω
1R 330 Ω
3TB (H)
3TB (F)
3TB (G)
7P 100Ω VOLT. RANGE
8P 250 Ω
2R 220Ω
1Z 9V
10P 5K INC
M
JOINT VOLTAGE REGULATOR
REFERENCE AND COMPARISON
CIRCUIT
3S7931JC200A4
UNIT GENERATOR
(TYPICAL FOR EACH GENERATOR)
T18(Δ)
T19(Δ)
P.T.
(6)
(7)
T1
15K(L-L)/115(L-L)
P.T. BURDEN
2.4VA/120VAC
(Δ)
Q-TABLE
T20
T21
Ø2
CT
(2)
(3)
T2
5000:5A
C.T. BURDEN
1.8VA/5A
Ø1 Ø2 Ø3
PHASE ROTATION
1-2-3
310Ω R12
20Ω R1
80Ω R2
RT-1
110Ω R3
368Ω R4
CR1
CR2
CR3 10V
CR4 10V
CR5 10V
1100Ω R5
517Ω R6
385Ω R7
CR6 10V
CR7 10V
CR8 13V
C1 .068 MFD
C2 100 MFD
R11 150Ω
R10 500Ω
R9 30Ω
R8 500Ω
C3 50MFD
– (1)
+ (12)
50-470121AWAA1
(GEI-88846 – WEST LYNN)

VAR TRANSDUCER (MOUNTED IN JVC CUBICLE)

NOTE: OUTPUT TERMINAL (12) WILL BE POSITIVE WITH RESPECT TO (1) FOR OVEREXCITED VARS WITH INSTRUMENT TRANSFORMER POLARITIES AS SHOWN.

WHEN THE GENERATOR IS SUPPLYING RATED VARS TO THE POWER SYSTEM, THE SIGNALS TO THE TRANSDUCER ARE 115V (POTENTIAL) AND I1. (REACTIVE COMPONENT ONLY – TOTAL CURRENT, AT RATED KVA, IS I2. THE TRANSDUCER IS CALIBRATED TO DELIVER 1.000MA INTO 2200Ω, (2.2V) WITH 500 VARS APPLIED. THUS THE OUTPUT CORRESPONDING TO GENERATOR RATED VARS WILL BE $2.2V \times \frac{115V \times I1}{500\ VA}$ = VOLTS

SEE CHART BELOW

	I2	I1	V
GEN. 1	4.27	1.86	.943
GEN. 2	4.27	1.86	.943
GEN. 3	4.27	1.86	.943

REVISIONS		PRINTS TO
1	REVISED McLaughlin APR 12 1978	3E H/
2	RGP / 8-14-80 GEN CORRECTIONS	

MADE BY 9/27/77
ISSUED OCT 6 1977
APPROVALS
DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT
WAYNESBORO, VA. LOCATION
44C308284
CONT ON SHEET 6 SH NO. 5

FF-809-F (2-77)
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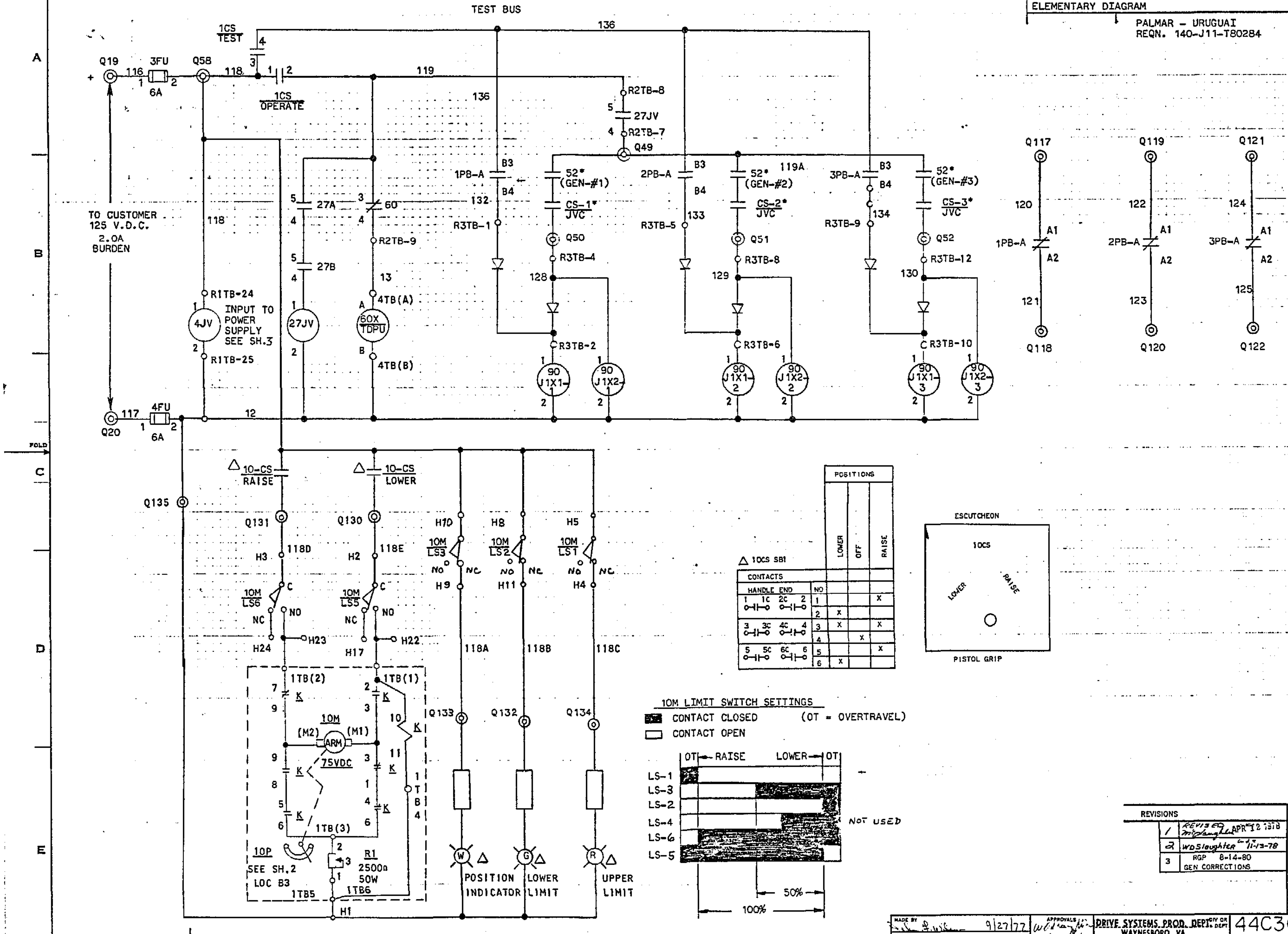
GENERAL ELECTRIC 44C308284
CONT ON SHEET 7 SH NO. 6

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:–				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
JOINT VOLTAGE CONTROL
FIRST MADE FOR 3S7931JC100A5
ELEMENTARY DIAGRAM

PALMAR – URUGUAI
REQN. 140-J11-T80284


TEST BUS
136
1CS TEST
1CS OPERATE
Q19
+
116
3FU
6A
Q58
118
119
TO CUSTOMER
125 V.D.C.
2.0A
BURDEN
R1TB-24
4JV
INPUT TO POWER SUPPLY SEE SH.3
R1TB-25
27A
27B
27JV
60
R2TB-9
13
4TB(A)
60X TDPU
4TB(B)
R2TB-8
27JV
R2TB-7
Q49
1PB-A
B3
B4
132
R3TB-1
52* (GEN-#1)
CS-1* JVC
Q50
R3TB-4
128
R3TB-2
90 J1X1-
90 J1X2-
2PB-A
133
R3TB-5
52* (GEN-#2)
119A
CS-2* JVC
Q51
R3TB-8
129
R3TB-6
90 J1X1-2
90 J1X2-2
3PB-A
134
R3TB-9
52* (GEN-#3)
CS-3* JVC
Q52
R3TB-12
130
R3TB-10
90 J1X1-3
90 J1X2-3
Q117
120
1PB-A
A1
A2
121
Q118
Q119
122
2PB-A
123
Q120
Q121
124
3PB-A
125
Q122
Q20
117
4FU
6A
12
Q135
10-CS RAISE
10-CS LOWER
Q131
118D
H3
10M LS6
C
NC
NO
H24
H23
Q130
118E
H2
10M LS5
H17
H22
1TB(2)
1TB(1)
K
10M
(M2)
ARM
(M1)
75VDC
1TB(3)
10P
SEE SH.2
LOC B3
R1
2500Ω
50W
1TB5
1TB6
1TB4
H1
H10
10M LS3
H9
118A
Q133
W
POSITION INDICATOR
H8
10M LS2
H11
118B
Q132
G
LOWER LIMIT
H5
10M LS1
H4
118C
Q134
R
UPPER LIMIT
ESCUTCHEON
10CS
LOWER
RAISE
PISTOL GRIP
10M LIMIT SWITCH SETTINGS
CONTACT CLOSED
CONTACT OPEN
(OT = OVERTRAVEL)
OT
RAISE
LOWER
LS-1
LS-3
LS-2
LS-4
LS-6
LS-5
NOT USED
100%
50%
FOLD

△ 10CS SB1

CONTACTS HANDLE END	NO	LOWER	OFF	RAISE
1 1C 2C 2	1			X
	2	X		
3 3C 4C 4	3	X		X
	4		X	
5 5C 6C 6	5			X
	6	X		

REVISIONS		PRINTS TO	
1	REVISED APR 12 1978		3E+
2	WDSlaughter 11-13-78		
3	RGP 8-14-80 GEN CORRECTIONS		

MADE BY 9/27/77
ISSUED OCT 6 1977
APPROVALS
DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT
WAYNESBORO, VA. LOCATION
44C308284
CONT ON SHEET 7 SH NO. 6

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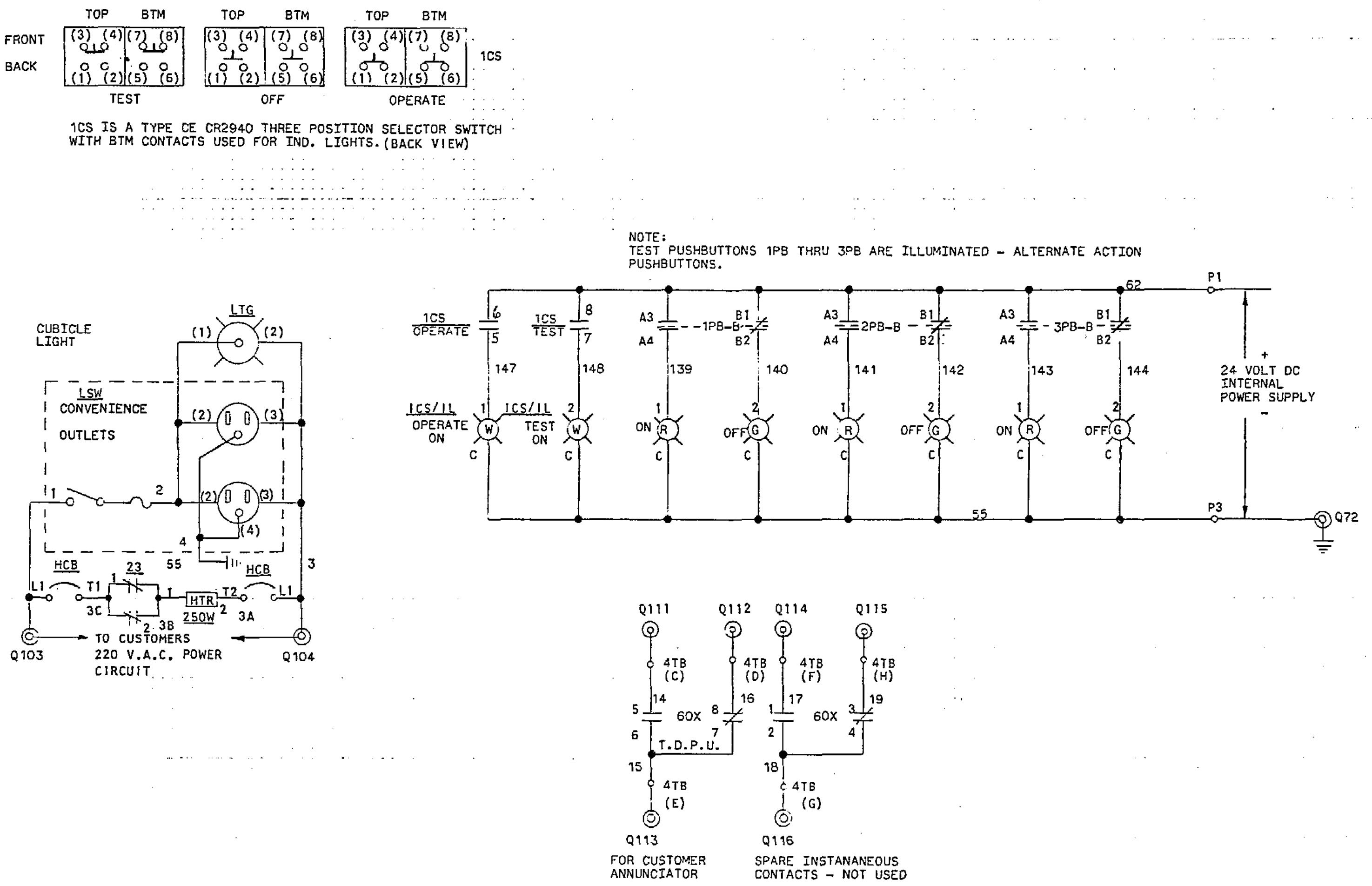
GENERAL ELECTRIC

44C308284
CONT ON SHEET 8 SH NO. 7

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS		
		FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
JOINT VOLTAGE CONTROL
FIRST MADE FOR 3S7931JC100A5
ELEMENTARY DIAGRAM

PALMAR - URUGUAI
REQN. 140-J11-T80284



TOP
BTM
FRONT
BACK
(3) (4) (7) (8)
(1) (2) (5) (6)
TEST
OFF
OPERATE
1CS
CUBICLE LIGHT
LTG
(1)
(2)
LSW
CONVENIENCE OUTLETS
(3)
(4)
1
2
3
4
55
L1
T1
T2
HCB
23
3C
2 3B
HTR
250W
3A
Q103
Q104
TO CUSTOMERS 220 V.A.C. POWER CIRCUIT
1CS OPERATE
1CS TEST
6
5
8
7
147
148
1CS/IL OPERATE ON
1CS/IL TEST ON
W
C
A3
A4
B1
B2
1PB-B
2PB-B
3PB-B
139
140
141
142
143
144
ON
OFF
R
G
62
P1
P3
+
24 VOLT DC INTERNAL POWER SUPPLY
-
Q72
Q111
Q112
Q114
Q115
4TB (C)
4TB (D)
4TB (F)
4TB (H)
14
16
17
19
60X
T.D.P.U.
15
18
4TB (E)
4TB (G)
Q113
Q116
FOR CUSTOMER ANNUNCIATOR
SPARE INSTANANEOUS CONTACTS - NOT USED

1CS IS A TYPE CE CR2940 THREE POSITION SELECTOR SWITCH WITH BTM CONTACTS USED FOR IND. LIGHTS. (BACK VIEW)

NOTE:
TEST PUSHBUTTONS 1PB THRU 3PB ARE ILLUMINATED - ALTERNATE ACTION PUSHBUTTONS.

REVISIONS		PRINTS TO	
1	REVISED 4 APR 1 [illegible] McLAUGHLIN	3EN1	
2	RGP 8-14-80 GEN CORRECTIONS		

MADE BY P.W. 9/27/77
ISSUED OCT 6 1977
APPROVALS
DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT
WAYNESBORO, VA. LOCATION
44C308284
CONT ON SHEET 8 SH NO. 7

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GENERAL ELECTRIC

44C308284
CONT ON SHEET 9 SH NO. 8

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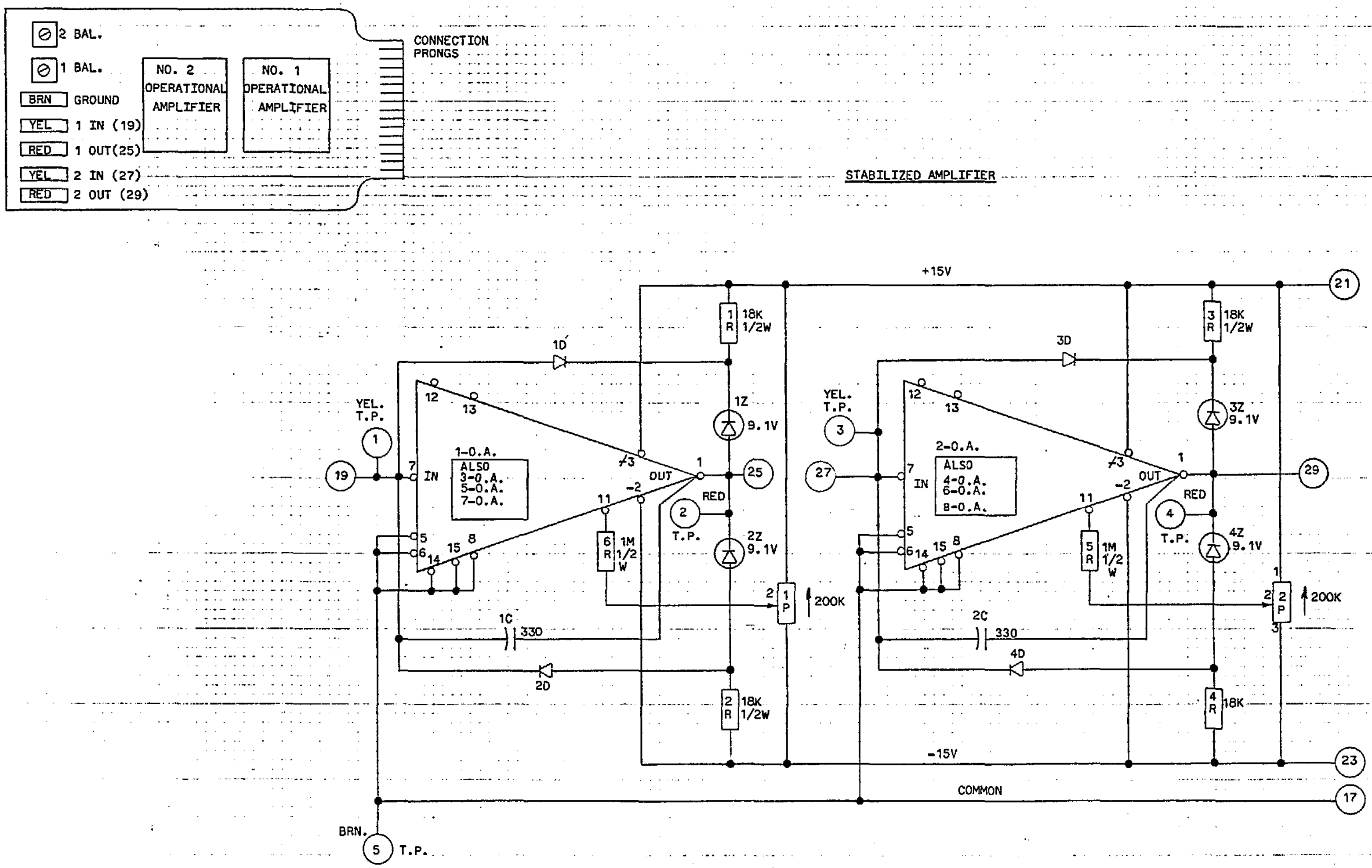
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
JOINT VOLTAGE CONTROL
FIRST MADE FOR 3S7931JC100A5
ELEMENTARY DIAGRAM

PALMAR – URUGUAI
REQN. 140-J11-T80284



2 BAL.
1 BAL.
BRN GROUND
YEL 1 IN (19)
RED 1 OUT(25)
YEL 2 IN (27)
RED 2 OUT (29)
NO. 2 OPERATIONAL AMPLIFIER
NO. 1 OPERATIONAL AMPLIFIER
CONNECTION PRONGS
STABILIZED AMPLIFIER
+15V
-15V
COMMON
YEL. T.P.
1
19
1-O.A.
ALSO 3-O.A. 5-O.A. 7-O.A.
IN
OUT
1D
1Z 9.1V
25
RED
2
T.P.
2Z 9.1V
6 R 1M 1/2 W
1 R 18K 1/2W
2 R 18K 1/2W
1 P 200K
1C 330
2D
YEL. T.P.
3
27
2-O.A.
ALSO 4-O.A. 6-O.A. 8-O.A.
3D
3Z 9.1V
29
4
4Z 9.1V
5 R 1M 1/2 W
3 R 18K 1/2W
4 R 18K
2 P 200K
2C 330
4D
21
23
17
BRN.
5
T.P.

NOTE

BALANCE ADJUSTMENTS SHOULD BE CENTERED FOR NORMAL OPERATION.

REVISIONS		PRINTS TO
1	RGP 8-14-80 GEN CORRECTIONS	3E#1

MADE BY J. Wilson 9/27/77
ISSUED OCT 6 1977
APPROVALS W. Clangh W. H. Apulen
DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.

44C308284
CONT ON SHEET 9 SH NO. 8

FF-809-F (2-77)
PRINTED IN U.S.A.

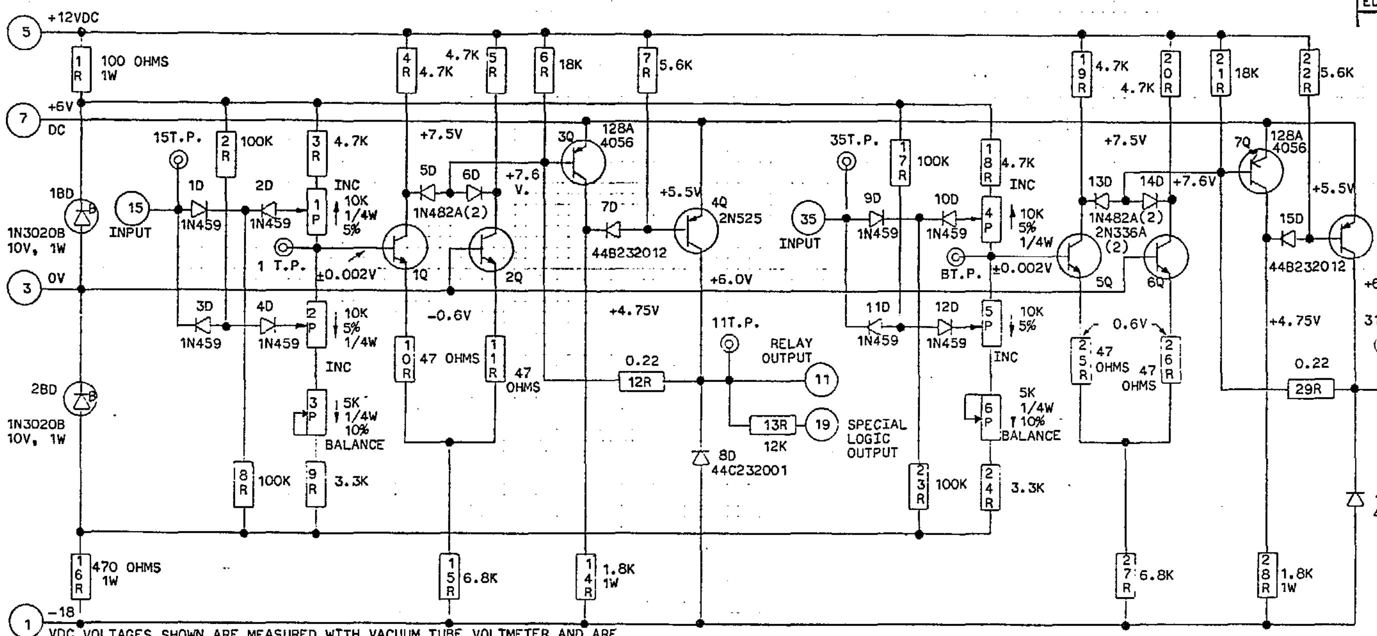
GENERAL ELECTRIC

44C308284 CONT ON SHEET 10 SH NO. 9

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

TITLE
JOINT VOLTAGE CONTROL
FIRST MADE FOR 3S7931JC100A5
ELEMENTARY DIAGRAM

LEVEL SENSER BOARD



5
+12VDC
1R 100 OHMS 1W
7
+6V DC
18D
1N3020B 10V, 1W
3
0V
28D
1N3020B 10V, 1W
16R 470 OHMS 1W
1
-18
15T.P.
15
INPUT
1D 1N459
2D 1N459
3D 1N459
4D 1N459
2R 100K
8R 100K
3R 4.7K
1P INC 10K 1/4W 5%
1 T.P.
±0.002V
2P 10K 5% 1/4W INC
3P 5K 1/4W 10% BALANCE
9R 3.3K
4R 4.7K
5R 4.7K
+7.5V
5D
6D
1N482A(2)
+7.6 V.
1Q
2Q
-0.6V
10R 47 OHMS
11R 47 OHMS
15R 6.8K
6R 18K
3Q 128A 4056
14R 1.8K 1W
7R 5.6K
7D 44B232012
+5.5V
4Q 2N525
+6.0V
+4.75V
12R 0.22
11T.P.
RELAY OUTPUT
11
13R 12K
19
SPECIAL LOGIC OUTPUT
8D 44C232001
35T.P.
35
INPUT
9D 1N459
10D 1N459
11D 1N459
12D 1N459
17R 100K
23R 100K
18R 4.7K INC
4P 10K 5% 1/4W
BT.P.
±0.002V
5P 10K 5% INC
6P 5K 1/4W 10% BALANCE
24R 3.3K
19R 4.7K
20R 4.7K
+7.5V
13D
14D
+7.6V
1N482A(2)
2N336A (2)
5Q
6Q
0.6V
25R 47 OHMS
26R 47 OHMS
27R 6.8K
21R 18K
7Q 128A 4056
28R 1.8K 1W
22R 5.6K
15D 44B232012
+5.5V
8Q 2N525
+60V
+4.75V
29R 0.22
31 T.P.
31
RELAY OUTPUT
30R 12K
39
SPECIAL LOGIC OUTPUT
16D 44B232001

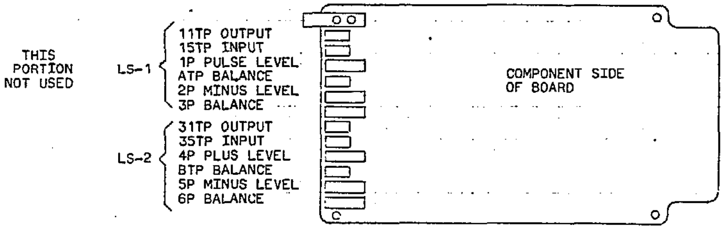
PALMAR – URUGUAI
REQN. 140–J11–T80284

PIN NO.	FUNCTION
1	-18V DC
3	0 VOLTS
5	+12V DC
7	+6V DC
9	NO CONNECTION
11	RELAY OUTPUT LS-1
13	NO CONNECTION
15	INPUT LS-1
17	NO CONNECTION
19	LOGIC OUTPUT LS-1
21	NO CONNECTION
23	NO CONNECTION
25	NO CONNECTION
27	NO CONNECTION
29	NO CONNECTION
31	RELAY OUTPUT LS-2
33	NO CONNECTION
35	INPUT LS-2
37	NO CONNECTION
39	LOGIC OUTPUT LS-2
41	NO CONNECTION

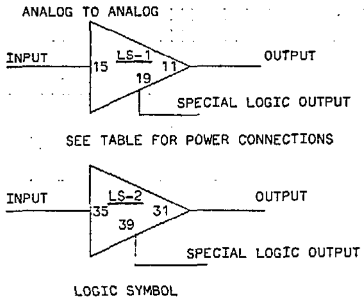
NOTE: LS-1 NOT USED IN THIS EQUIPMENT. IT MAY BE USED AS A SPARE IN PLACE OF LS-2 BY CHANGING CONNECTIONS.

VDC VOLTAGES SHOWN ARE MEASURED WITH VACUUM TUBE VOLTMETER AND ARE REFERRED TO 0 VOLTS. VALUES ARE TYPICAL AND MAY DEVIATE ±10% FROM THOSE SHOWN.

UNLESS OTHERWISE SPECIFIED:
RESISTANCE VALUES ARE 5%
RESISTANCE WATTAGE RATING IS 1/2 WATT



THIS PORTION NOT USED
LS-1
11TP OUTPUT
15TP INPUT
1P PULSE LEVEL
ATP BALANCE
2P MINUS LEVEL
3P BALANCE
LS-2
31TP OUTPUT
35TP INPUT
4P PLUS LEVEL
BTP BALANCE
5P MINUS LEVEL
6P BALANCE
COMPONENT SIDE OF BOARD



ANALOG TO ANALOG
INPUT
15
LS-1
11
19
OUTPUT
SPECIAL LOGIC OUTPUT
SEE TABLE FOR POWER CONNECTIONS
INPUT
35
LS-2
31
39
OUTPUT
SPECIAL LOGIC OUTPUT
LOGIC SYMBOL

DESCRIPTION:

THIS UNIT CONTAINS TWO IDENTICAL SECTIONS. THE INPUT TO THE "A" SECTION IS INTO A DIODE SWITCH 2D AND 4D WHICH CONNECTS THE INPUT SIGNAL TO THE BASE OF 1Q WHEN THE INPUT EXCEEDS A LEVEL IN EXCESS OF THAT AT THE SLIDERS OF 1P AND 2P. INPUT SWITCH BIAS IS PROVIDED BY DIODES 1D AND 3D. THE INPUT SIGNAL APPLIED TO THE BASE OF 1Q IS AMPLIFIED AND SWITCHED TO THE BASE OF 3Q BY MEANS OF DIODES 5D AND 6D. TRANSISTOR 3Q DRIVES 4Q TO OPERATE THE RELAY. SNAP ACTION IS INSURED BY FEEDBACK THROUGH RESISTOR 12R. OPERATION OF THE "B" SECTION IS SIMILAR.

ADJUSTMENTS:

WITH NO INPUT SIGNAL THE OUTPUT IS +6 VOLTS DC AND THE ASSOCIATED RELAY IS ENERGIZED. UPON APPLICATION OF AN APPROPRIATE INPUT SIGNAL THE RELAY IS DE-ENERGIZED THE INPUT OPERATING LEVEL MAY BE ADJUSTED FOR POSITIVE GOING SIGNALS BY MEANS OF 1P/4P AND FOR NEGATIVE GOING SIGNALS BY MEANS OF 2P/5P OVER THE RANGE OF ±0.5 TO ±5.0 VOLTS DC.

BALANCE CONTROLS 3P AND 6P SERVE TO BALANCE THE SWITCHED INPUT AT A.T.P. & B.T.P. TO ZERO WITH ZERO INPUT.

THE SIGNAL INPUT IMPEDANCE IS APPROXIMATELY 100K OHMS AND A MAXIMUM OF ±175V DC MAY BE APPLIED AT THE INPUT.

REVISIONS		PRINTS TO
1	RGP 8-14-80 GEN CORRECTIONS	3EH/

MADE BY 9/27/77
ISSUED OCT 6 1977
APPROVALS
DRIVE SYSTEMS PROD. DEPT
WAYNESBORO, VA.
LOCATION

44C308284 CONT ON SHEET 10 SH NO. 9

FF-808-F (2-77) PRINTED IN U.S.A.

GENERAL ELECTRIC

44C308284
CONT ON SHEET 11 SH NO. 10

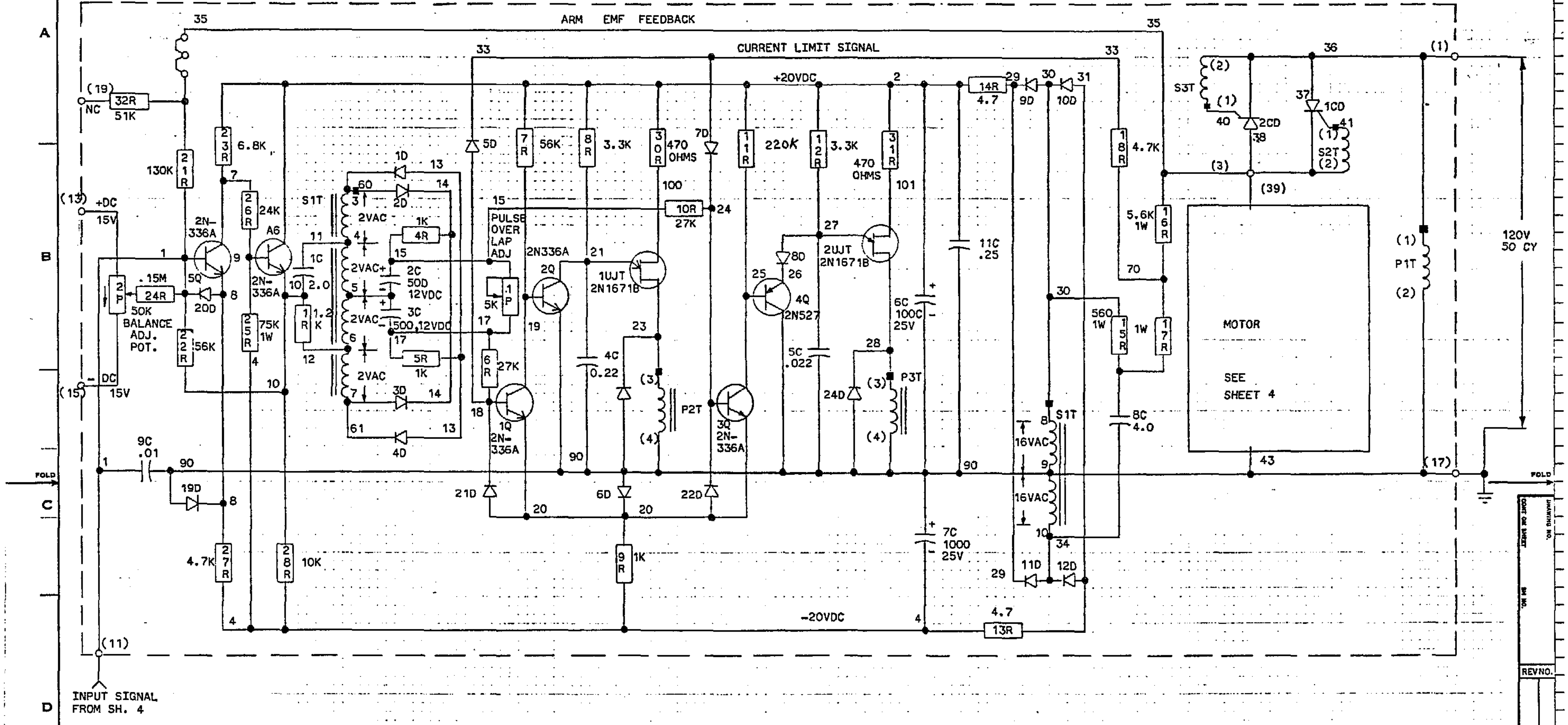
TITLE
JOINT VOLTAGE CONTROL

FIRST MADE FOR 3S7931JC100A5

ELEMENTARY DIAG. PALMAR - URUGUAI
REQN. 140-J11-T80284

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

HALF WAVE REVERSING MOTOR CONTROL
WITH CURRENT LIMIT (3/4 HP MAX.)



ARM EMF FEEDBACK
CURRENT LIMIT SIGNAL
+20VDC
-20VDC
35
33
(19) NC
32R 51K
130K 21R
(13) +DC 15V
(15) -DC 15V
2P 50K BALANCE ADJ. POT.
.15M 24R
20D
22R 56K
2N-336A 5Q
23R 6.8K
26R 24K
A6 2N-336A
25R 75K 1W
10
1C 2.0
1.2K 1R
S1T
2VAC
2VAC+
2VAC-
60
61
1D
2D
3D
4D
4R 1K
5R 1K
2C 50D 12VDC
3C 500,12VDC
PULSE OVER LAP ADJ
5K 1P
6R 27K
1Q 2N-336A
5D
7R 56K
2Q 2N336A
8R 3.3K
4C 0.22
1UJT 2N1671B
30R 470 OHMS
10R 27K
7D
P2T
3Q 2N-336A
11R 220K
4Q 2N527
8D
5C .022
12R 3.3K
2UJT 2N1671B
31R 470 OHMS
24D
P3T
6C 100C 25V
11C .25
14R 4.7
9D
10D
16VAC
11D
12D
7C 1000 25V
13R 4.7
18R 4.7K
16R 5.6K 1W
15R 560 1W
17R 1W
8C 4.0
S3T
2CD
1CD
S2T
MOTOR
SEE SHEET 4
P1T
120V 50 CY
9C .01
19D
27R 4.7K
28R 10K
9R 1K
21D
6D
22D
(11)
INPUT SIGNAL FROM SH. 4
(1)
(17)
(3)
(39)

REVISIONS		PRINTS TO
1	RGP 8-14-80 GEN CORRECTIONS	3EH.

MADE BY [illegible] 9/27/77
ISSUED OCT 8 1977
APPROVALS W.A. [illegible]

DRIVE SYSTEMS PROD. DEPT.
WAYNESBORO, VA.

44C308284
CONT ON SHEET 11 SH NO. 10

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GENERAL ELECTRIC

44C308284
CONT ON SHEET FNL SH NO. 11

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING:—				
APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
	✓	±	±	±

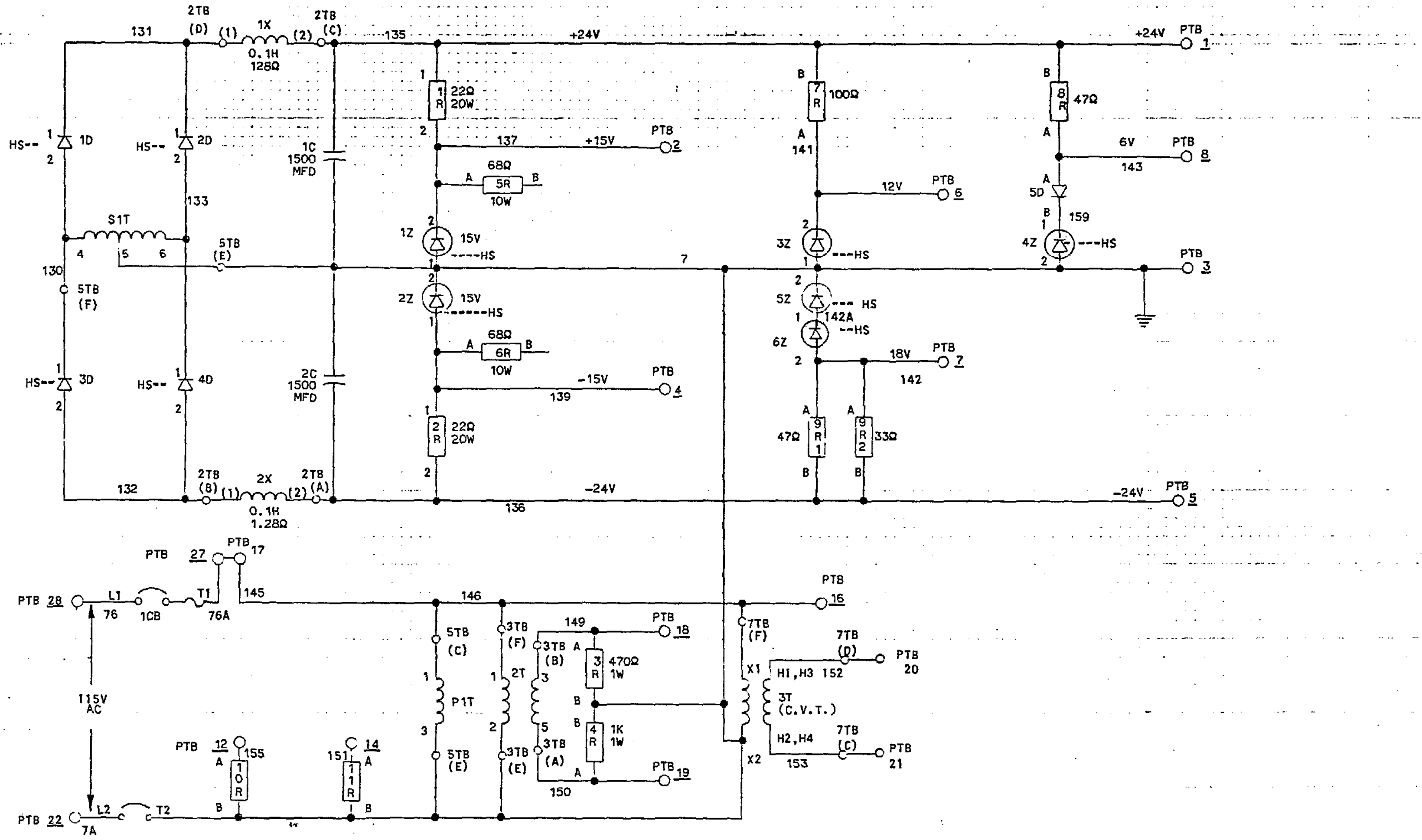
TITLE
JOINT VOLTAGE CONTROL

FIRST MADE FOR 3S7931JC100A5

ELEMENTARY DIAGRAM

PALMAR – URUGUAI
REQN. 140-J11-T80284

POWER SUPPLY CHASSIS



131
2TB (D)
(1)
1X
0.1H
128Ω
(2)
2TB (C)
135
+24V
+24V
PTB 1
1 R 22Ω 20W
1
2
137
+15V
PTB 2
68Ω
A 5R B
10W
1Z
15V
---HS
2
1
2Z
15V
------HS
68Ω
A 6R B
10W
-15V
139
PTB 4
2 R 22Ω 20W
1
2
1C 1500 MFD
2C 1500 MFD
HS-- 1D
HS-- 2D
HS-- 3D
HS-- 4D
1
2
133
S1T
4
5
6
130
5TB (F)
5TB (E)
7
132
2TB (B)
2X
0.1H
1.28Ω
2TB (A)
136
-24V
-24V
PTB 5
B
7 R 100Ω
A
141
12V
PTB 6
3Z
---HS
5Z
--- HS
142A
6Z
--HS
18V
PTB 7
142
A
9 R 1 47Ω
B
9 R 2 33Ω
B
8 R 47Ω
6V
PTB 8
143
5D
159
4Z
---HS
PTB 3
PTB 27
PTB 17
PTB 28
L1
76
1CB
T1
76A
145
146
PTB 16
5TB (C)
P1T
1
3
5TB (E)
3TB (F)
2T
1
2
3TB (E)
149
3TB (B)
3
5
3TB (A)
3 R 470Ω 1W
4 R 1K 1W
PTB 18
PTB 19
150
7TB (F)
X1
X2
3T (C.V.T.)
H1,H3 152
H2,H4
153
7TB (D)
PTB 20
7TB (C)
PTB 21
115V AC
PTB 12
155
1 0 R
151
PTB 14
1 1 R
A
B
PTB 22
L2
T2
7A

NOTE: (1) THIS "STANDARD" POWER SUPPLY CONTAINS CIRCUITS AND FEATURES NOT USED WITH JVC EQUIPMENT 3S7931JC100A5. SPECIFICALLY NO CONNECTIONS ARE MADE TO TERMINALS 11, 12, 14, 15, 17, 20, 21, 23, 24, 25 OF PTB, THUS THESE CIRCUITS IN POWER SUPPLY ARE NOT USED. TERMINALS 16, 17, AND 27 OF PTB ARE JUMPERED BY WIRE NO. 156 TO BYPASS CONTACT OF UNUSED RELAY. TERMINALS 3 AND 22 GO TO GROUND.

(2) TIE AND TAPE LEAD 2T(4)

REVISIONS		PRINTS TO	
1	REVISED McLaughlin APR 12 1978		3EHI
2	RGP 8-14-80 GEN CORRECTIONS		

MADE BY	APPROVALS		
W. Wilson 9/27/77 ISSUED OCT 6 1977		DRIVE SYSTEMS PROD. DEPT. DIV OR DEPT WAYNESBORO, VA. LOCATION	44C308284 CONT ON SHEET FNL SH NO. 11

FF-809-F (2-77)
PRINTED IN U.S.A.

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC